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Paleomagnetic evidence for rotation in the Precordillera of northern Chile: structural constraints and implications for the evolution of the Andean forearc

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ABSTRACT

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Palaeomagnetic results are reported for two late Mesozoic to early Cenozoic continental sedimentary sequences from the Precordillera (Cordillera de Domeyko) of northern Chile. Comparison of isolated components of magnetisation from the Late Cretaceous–Palaeocene Purilactis Formation and the unconformably overlying Oligo-Miocene Paciencia Group with reference poles for South America indicate that significant post-Palaeocene and post-Lower Miocene clockwise rotation has taken place.

A structural study of the area has revealed the presence of a number of thrust sheets that were periodically active throughout the Cenozoic. Stratigraphic, structural and palaeomagnetic data indicate that thrust sheet movement took place following deposition of both the Purilactis Formation (?lower Eocene Incaic Orogeny) and the Paciencia Group (?lowest Miocene Peheunche Orogeny) and resulted in the folding, uplift and clockwise rotation of the two sequences. Differences in the amount of rotation between the two units reflect incremental movement of the thrust sheets, 12° of rotation took place prior to deposition of the Paciencia Group followed by 17° of post-Lower Miocene clockwise rotation.

This study demonstrates that palaeomagnetically detected rotation in the Precordillera of northern Chile, 250 km inland of the Peru–Chile trench, is related to the periodic movement of an easterly propagating thin-skinned thrust front. Between 22°30′ and 23°30′S the north Chilean forearc contains areas of non-rotation, thin-skinned compressional related rotations and large, block fault rotations indicating that the simplistic models previously presented to account for forearc rotation in the Central Andes should be modified. In particular, evidence implies that rotation could be related to a number of tectonic events and may not simply be due to one deformation event.

Introduction

Palaeomagnetic studies of the Central Andean Orogen have revealed both concordant and discordant directions when compared to equivalently aged cratonic data for South America (e.g., Palmer et al., 1980; May and Butler, 1985; Beck et al., 1986; Jesinkey et al., 1987; Beck, 1988). The data from the Andean forearc can essentially be split into two geographic areas: (1) north of 19°S where rotation is in an anticlockwise sense; and (2) south of 19°S where clockwise rotations and areas of non-rotation have been identified. The systematc variation in the direction of rotations around the Arica Deflection (Fig. 1) has lent support to the idea of oroclinal bending (Heki et al., 1983; Isacks, 1988; McFadden et al., 1990) first proposed by Carey (1955, 1958). However, marked variations in the amount of rotation within these groupings suggests that rotation is related to localised, systematic block rotations

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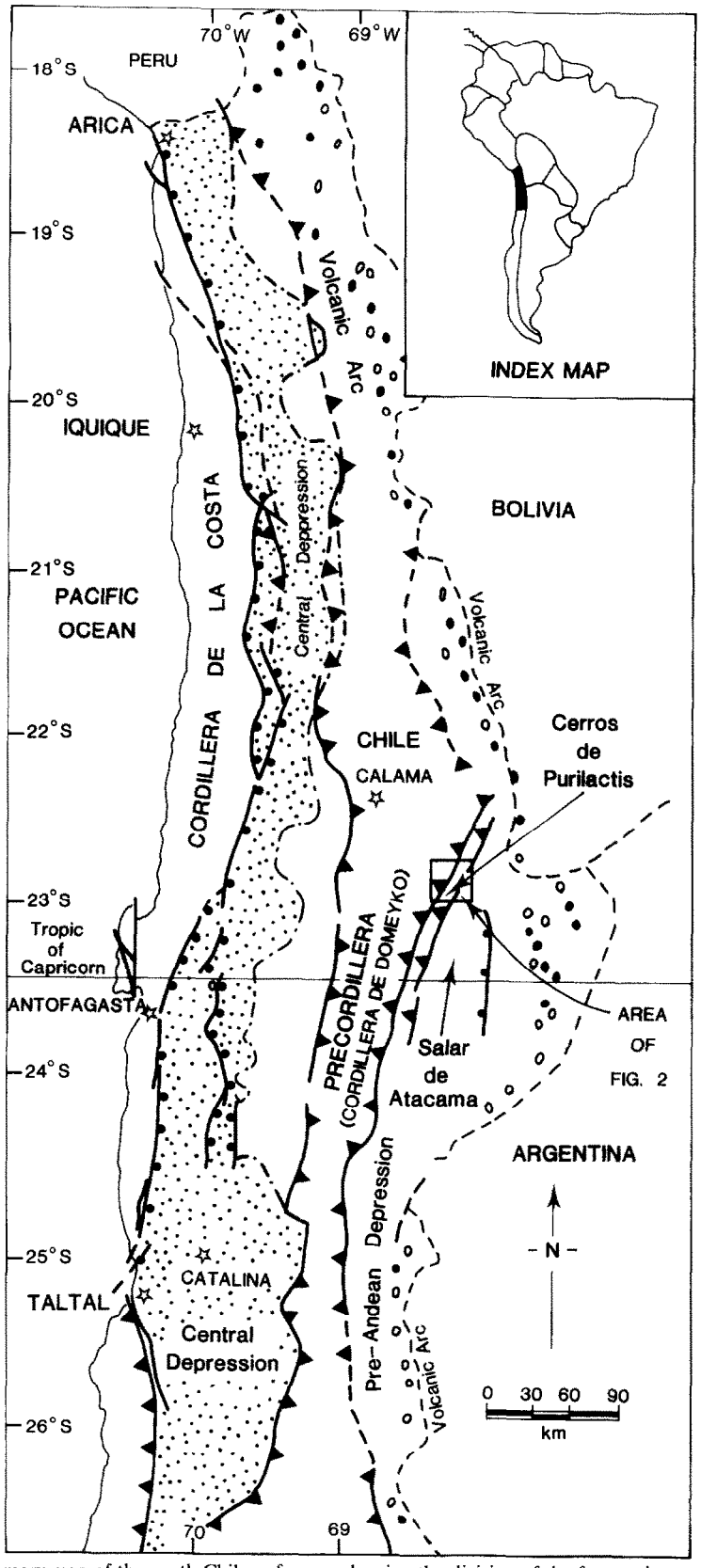


Fig. 1. Geological summary map of the north Chilean forearc showing the division of the forearc into morphotectonic units.

due to either strike-slip faulting generated by oblique plate convergence (Beck, 1988) or extension within the forearc (Hartley et al., 1988a).

Here we present palaeomagnetic data for Lower Tertiary rocks from the Precordillera of northern Chile. The Precordillera was chosen as a study area in order to establish whether rotation (which has largely been documented from the coastal area of the Andean Orogen) extended inland and if so, how it was related to the rotation which has been documented from the Cordillera de la Costa at the same latitude (Hartley et al., 1988a). This approach would allow an understanding of the tectonic mechanisms required to drive rotation within the forearc and should allow some constraints to be placed on the orocline hypothesis.

Geology

The Precordillera of northern Chile is a narrow mountain range up to 4500 m high located within the Andean forearc between the Central and Pre-Andean Depressions (Fig. 1). The study area comprises the northern part of the Cordillera de Domeyko and forms the northwestern margin of the Salar de Atacama (Fig. 2).

Stratigraphy

The stratigraphy of the area is illustrated in Figure 3 and was principally established by Brüggen (1942, 1950), Dingman (1963, 1967) and Marinović and Lahsen (1984). It comprises a local basement of Permo-Triassic lavas and sediments (the Tuina, Agua Dulce and El Bordo Formations) unconformably overlain by the Tonel Formation (Fig. 3). The Tonel Formation is a continental sequence comprising fine-grained sandstones, mudstones and evaporites intruded by hornblende rich basic dykes. The age of the formation is thought to be largely Cretaceous as it is conformably overlain by the Purilactis Formation (see below).

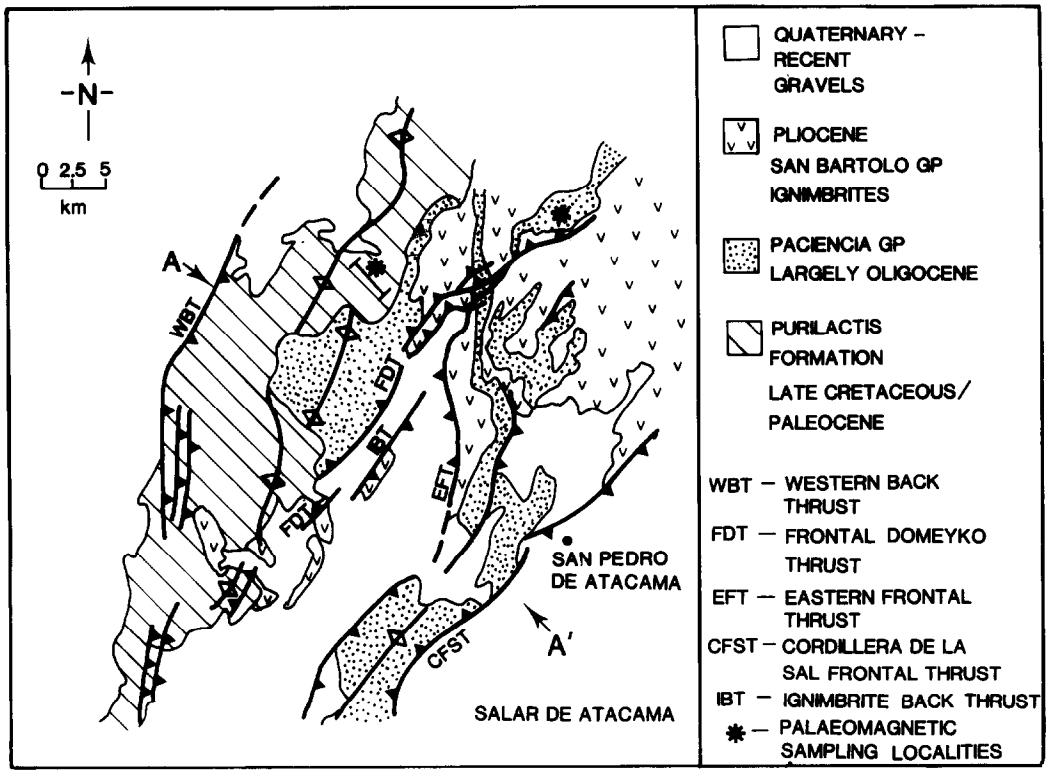


Fig. 2. Detailed geological map of the study area within the Precordillera, based on Jolley et al. (1990). A-A′ corresponds to the line of section illustrated in Fig. 4.

The Purilactis Formation is 3500 m thick and comprises a variety of alluvial and aeolian facies (Hartley et al., 1988b; Flint et al., 1989). Recent dating of a lava within the Formation indicates a Palaeocene age for the upper part of the section (Flint et al., 1989). An angular unconformity separates the Purilactis Formation from the overlying continental alluvial sediments of the 2000-m-thick Paciencia Group (Hollingworth and Rutland, 1968; Flint, 1985a). The group has been assigned an Oligo-Miocene age based on a 28 ± 6 Ma radiometric date from an ignimbrite within the upper part of the Group (Travisany, 1978) and a 17 ± 2 Ma age from an unconformably overlying lava flow (Ramírez, 1979). The lava forms part of the extensive, ignimbrite-dominated San Bartolo Group (Fig. 3; Hollingworth and Rutland, 1968; Ramírez, 1979). The stratigraphy of the area is completed by a series of Quaternary-Recent gravels and playa facies interbedded with thin volcanic ash beds (Jolley et al., 1990).

Structure

The structure of the Precordillera of northern Chile has been documented by Jolley et al. (1990), consequently, only a brief description is presented here. The study area is dominated by a series of major easterly and westerly verging thrust sheets which have been periodically active throughout the Tertiary (Fig. 4). Two major bounding faults have been identified (Fig. 4), the western back thrust (WBT) and the frontal Domeyko thrust (FDT) (Jolley et al., 1990). Where observed they are steep dipping (50–60°) thrust planes, displaying dominantly dip-slip movement.

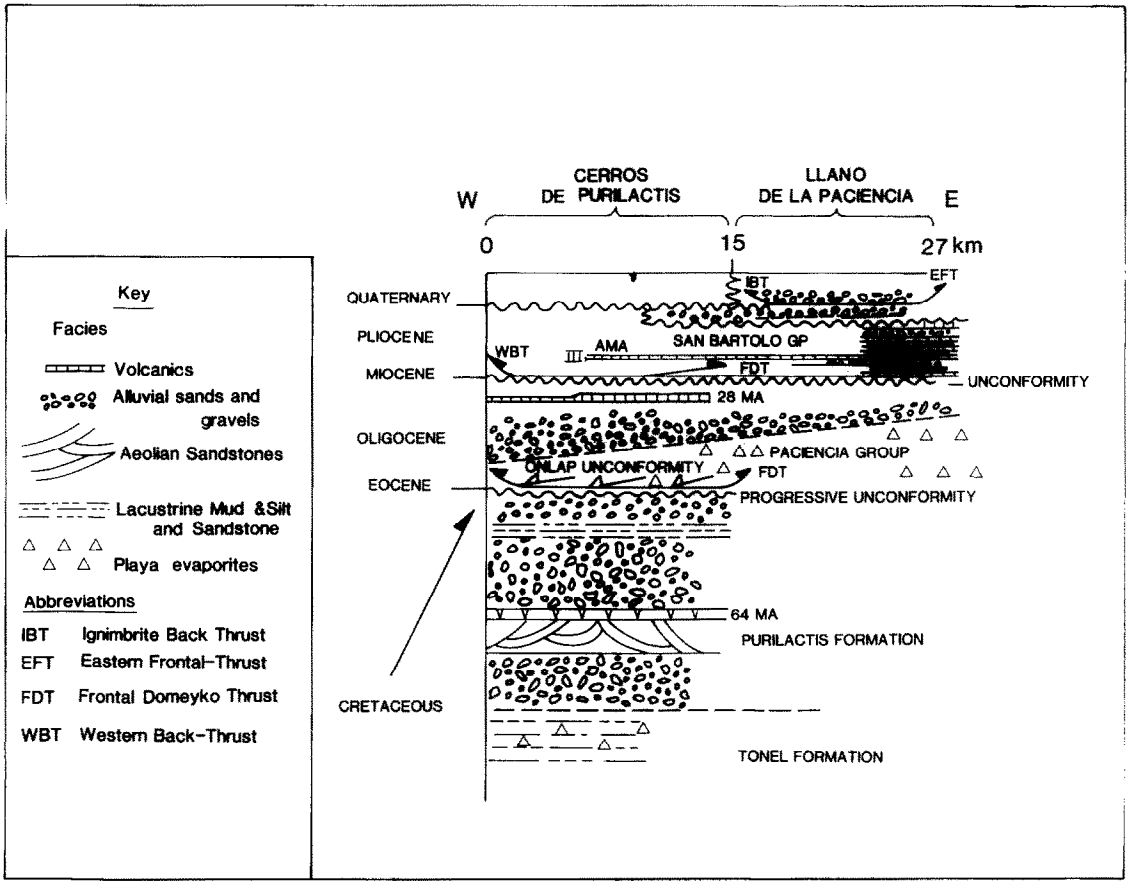


Fig. 3. A stratigraphic evolution diagram for the study area from the Cretaceous to Recent, illustrating facies changes, major unconformities and the timing and position of active faulting within the stratigraphic sequence. Modified after Jolley et al. (1990).

However, the WBT also displays dip surfaces and conjugate fractures recording a component of oblique and strike-slip movement with a dominantly sinistral sense. Dating thrust movements suggests that they propagated from west to east from the Precordillera into the Salar de Atacama, where the fault style becomes increasingly thin-skinned (Jolley et al., 1990).

The beds of the Purilactis Formation and Paciencia Groups have a similar structural strike. Strata of the Purilactis Formation are more steeply dipping and regionally are folded into a broad syncline bounded to the west by the WBT and to the east by the FDT. Locally, this simple geometry is complicated by east- and west-verging folds and thrusts. The more gently dipping Paciencia Group strata and the unconformity between the two units are folded into a shallow asymmetric syncline. The gentle easterly dip of the western limb gives way to abrupt steepening on the eastern limb of the syncline as the beds become deformed in the hanging wall of the FDT. The Purilactis Formation is interpreted to be part of a large 'pop-up' structure created during inversion of the Tonel/Purilactis basin-fill along older, reactivated basin-forming faults such as the FDT. The FDT was subsequently reactivated during uplift of the Paciencia Group.

Palaeomagnetic results

Oriented hand samples and drill cores were collected from the Purilactis Formation and the Paciencia Group (Fig. 2) from which 4–7 specimen cores were prepared in the laboratory. Specimens were treated using stepwise thermal and alternating field (AF) demagnetisation techniques and natural remnant magnetisation (NRM) mea-

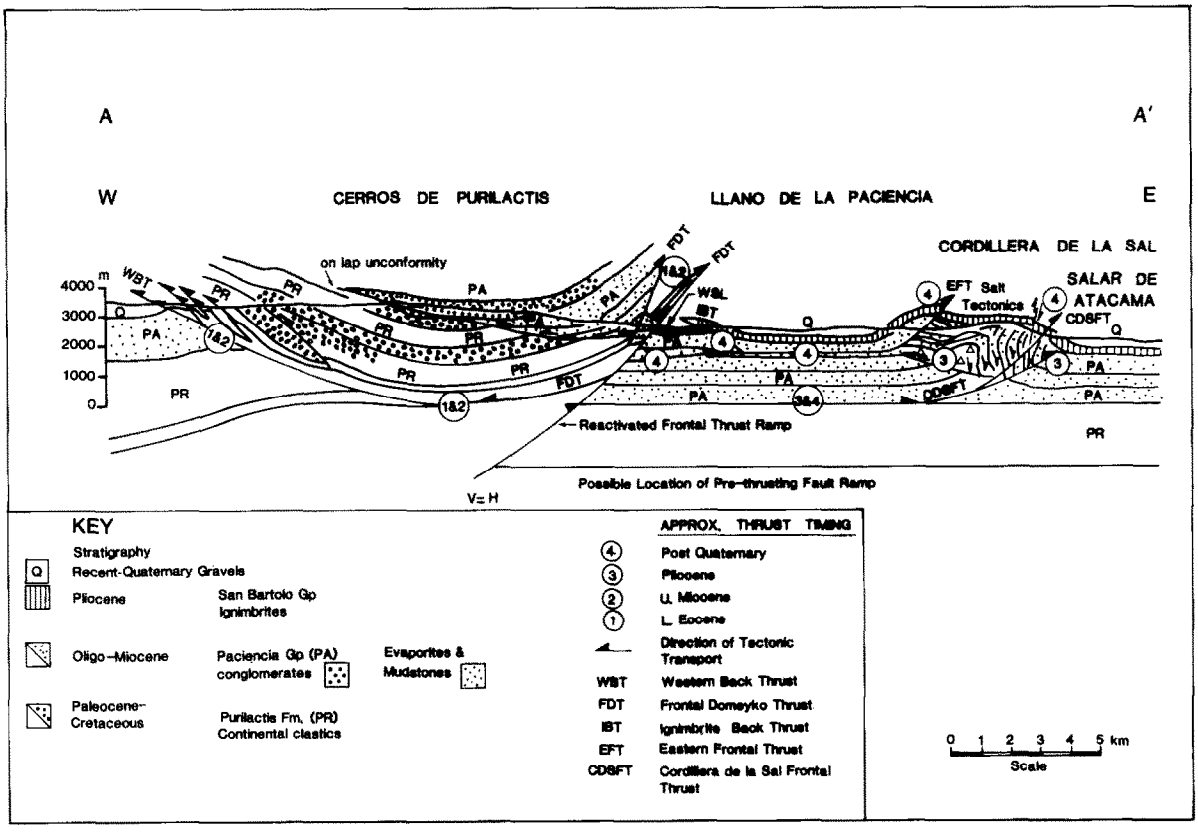


Fig. 4. Structural cross-section through the Cerros de Purilactis, Llano de la Paciencia and Cordillera de la Sal. The predominant direction of tectonic transport is to the east with minor amounts of westerly directed thrusting. From Jolley et al. (1990).

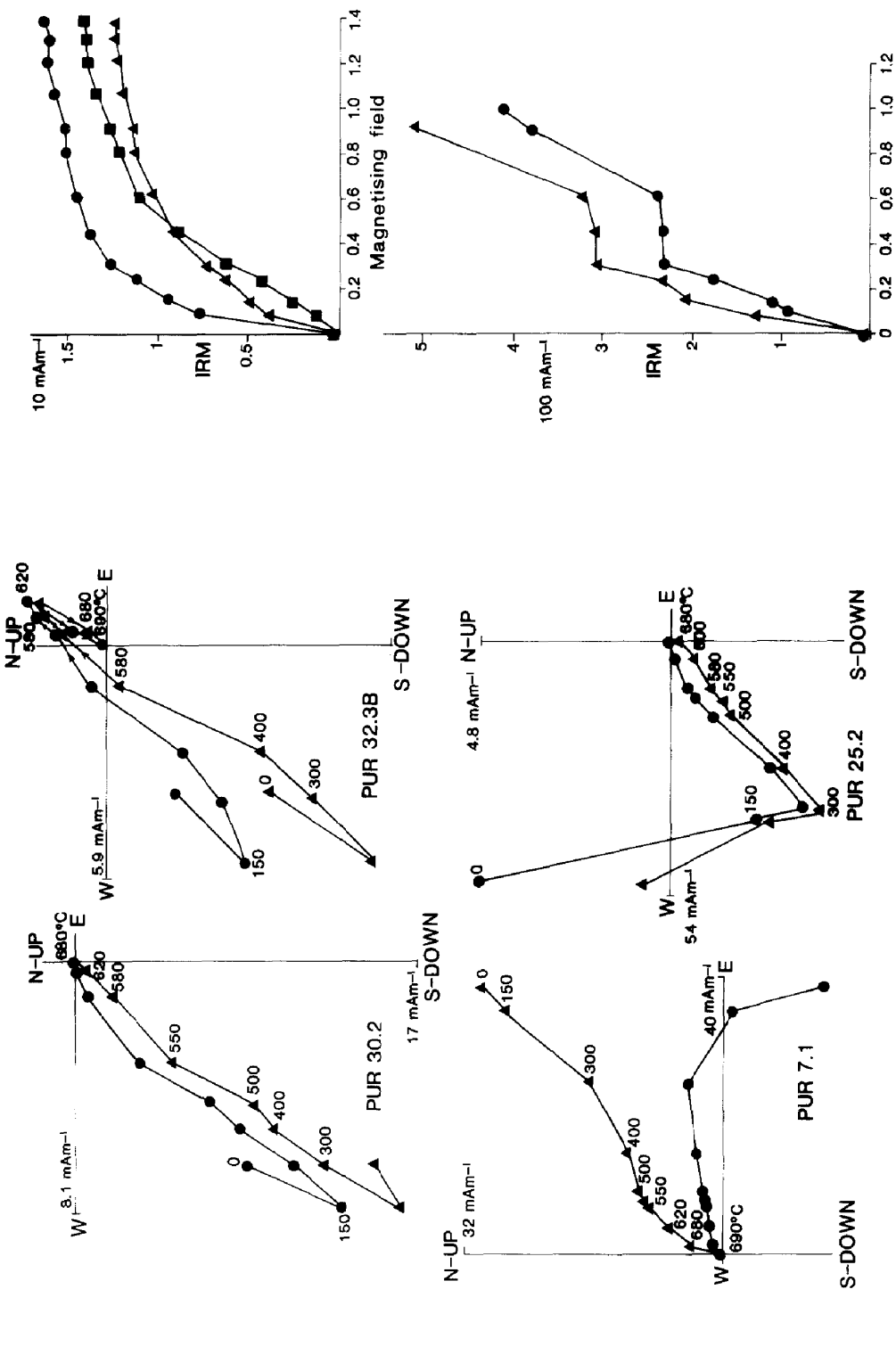


Fig. 5. Representative Zijderveld diagrams (circles = projection into horizontal plane, triangles = projection into vertical plane) and IRM acquisition curves for the Purilactis Formation. Numbers adjacent to data points are thermal demagnetisation temperatures (°C).

sured using a two-axis croyogenic magnetometer at the Nuffield Laboratory, Newcastle-upon-Tyne [see Hartley et al. (1988a) for further details]. Isolated components of magnetisation were extracted using principal component analysis (Kirschvink, 1980) to calculate the least-squares best-fitting line with a maximum angular deviation of less than 10°.

Purilactis Formation

Fourteen sites ($N = 69$) were collected from the eastern limb of the Purilactis syncline (Fig. 2). Unfortunately, sampling was restricted to the eastern limb, as the western limb is largely inaccessible, poorly exposed and structurally complex, comprising a series of thrust-bounded slivers of strata (Jolley et al., 1990). However, minor folds in the sample area allowed some constraint to be placed on the age of magnetisation (see Table 1). Sampling was restricted to fine- to coarse-grained sandstones interbedded with conglomerates from a section lying stratigraphically above the 64 Ma lava dated by Flint et al. (1989).

Samples were subject to thermal demagnetisation up to 680°C as preliminary AF demagnetisation showed that some samples retained up to 60% of the initial NRM at 100 mT. Stepwise demagnetisation revealed two categories of magnetisation: (1) sites which contain single components of magnetisation up to 680°C with directions either in the southwestern quadrant with a positive inclination (reversed) or in the northeastern quadrant with negative inclination (normal); and (2) sites which contain a low- to intermediate-temperature reversed component (0–550°C) and a high-temperature normal component (550–680°C) approximately antiparallel to the low-intermediate component. Components of magnetisation extracted by vector analysis over these

TABLE 1

Components of NRM for the Purilactis Formation. Statistics after Fisher (1953)

Site	N	k	D (°)	I (°)	α_{95}	Range (°)	Tilt	k	D (°)	I (°)	α_{95}
PUR2	6	5	352	−37	32.2	500–620	*	10	20	−26	21.8
PUR7	5	9	71	−15	26.4	500–620	038/33E	17	58	−30	19.3
PUR23	5	13	29	−71	22.1	550–680	163/18W	12	49	−55	21.7
PUR23R	5	11	190	+47	24.2	300–580	163/18W	23	203	+37	16.5
PUR24	6	24	215	+41	14.0	400–620	163/18W	44	222	+26	10.2
PUR25	6	40	204	+48	10.8	300–580	163/18W	40	215	+35	10.7
PUR26R	5	11	229	+52	24.2	400–550	147/22W	14	231	+30	21.5
PUR26	4	11	23	−39	23.4	550–680	147/22W	18	30	−21	20.8
PUR28	7	9	262	+73	20.9	500–620	143/35W	12	244	+40	18.5
PUR29R	5	5	210	+50	37.7	300–550	143/20W	18	216	+33	18.8
PUR29	4	11	30	−57	29.2	550–680	143/20W	12	35	−37	14.0
PUR30	3	30	183	+31	23.1	300–550	211/28W	999	203	+40	3.8
PUR32R	7	134	233	+33	5.2	150–500	253/22N	109	249	+38	5.6
PUR35	3	24	188	−6	26.1	300–580	248/34N	22	191	+23	26.5
PUR36	6	18	220	+44	16.5	300–580	243/40N	18	262	+45	16.5
Rev	10	11	209	+44	15.7			18	223	+37	11.8
Normal	5	7	31	−48	30.7			20	37	−34	17.6
Mean	15	10	30	−45	12.9			19	41	−36	8.9
Pole Position: 52°S 197°E.											

N = number of specimens prepared from oriented block samples; k = best estimate of Fisher precision parameter; I(°) and D(°) are site-mean inclination and declination, respectively; α_{95} = radius of cone of 95% confidence about the site-mean direction; range (°C) = temperature range over which component analysis was undertaken. Samples comprising PUR2 were taken from within 100 m of each other but have different tectonic corrections. R = reversed-polarity component extracted from site; tilt = tectonic correction referring to strike and direction of dip. The second set of corrections listed in the table have been structurally corrected.

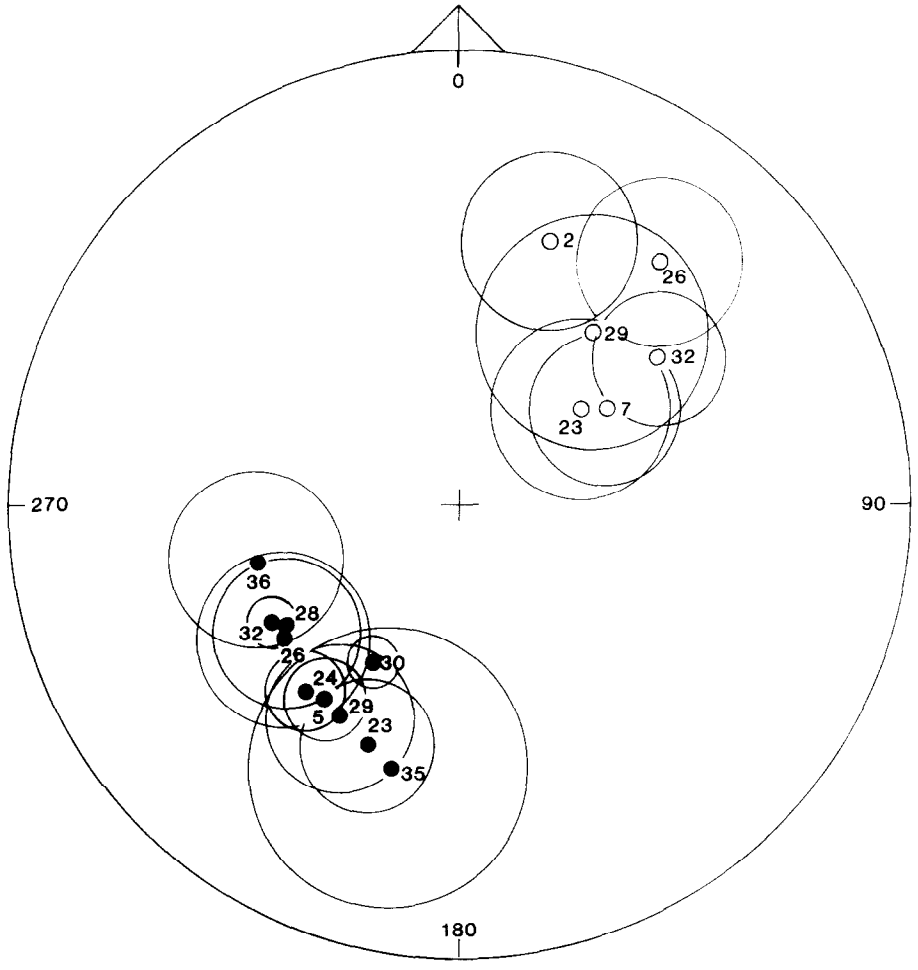


Fig. 6. Equal-area stereographic projection, site-mean NRM directions and α_{95} confidence limits for the Purilactis Formation, keyed to Table 1. Open data points are in the upper hemisphere projection; filled data points are in the lower hemisphere.

temperature ranges are presented in Table 1. Representative Zijderveld diagrams are shown in Figure 5. Site mean component directions and confidence limits are shown in Figure 6.

Isothermal remanent magnetisation (IRM) acquisition and reflected light petrography indicate that coarse-grained detrital magnetite and haematite are present (Fig. 5). Fine-grained pigmentary haematite also occurs as an alteration product of clasts and within the matrix of the sandstones.

The presence of antiparallel components of magnetisation within individual sites and the increased grouping of the data (Table 1) following differing bedding corrections suggest that magnetisation of the Purilactis Formation occurred relatively rapidly after deposition. This inference is also supported by the fact that the pole position for the formation differs significantly from that of the overlying Paciencia Group (see below). Thus, magnetisation of the Purilactis Formation took place prior to deposition of the Paciencia Group and is therefore pre-Oligocene and probably Palaeocene in age.

Paciencia Group

Thirteen sites ($N = 63$) were collected from medium- and coarse-grained sandstone horizons throughout the lower part of the Paciencia Group (Artolla Member of Flint, 1985a). Ten sites displayed essentially univectorial behaviour during

TABLE 2

Extracted components of magnetisation from the Paciencia Group

Site	N	k	D (°)	I (°)	α_{95}	Range (°)	Tilt	k	D (°)	I (°)	α_{95}
SBT1	3	33	277	−28	21.8	500–580	213/55W	32	223	−67	22.0
SBT2	6	19	301	−33	15.6	0–550	213/55W	71	263	−87	8.0
SBT4	6	83	356	−31	7.4	0–550	033/50E	109	346	+4	6.4
SBT5	5	18	254	−40	18.6	150–400	163/60W	19	69	−80	18.0
SBT6 *	3	140	188	+15	10.5	300–680	253/40N	143	202	+49	10.3
SBT8 *	3	228	190	−6	16.6	0–680	003/78E	248	190	+6	15.9
SBT10R *	6	19	190	−12	15.9	300–550	253/40N	41	192	+24	10.5
SBT10	6	27	355	−55	13.1	0–400	253/40N	27	112	−81	13.1
SBT13 *	5	14	23	−14	21.1	400–550	249/28N	18	32	−33	18.6
SBT14 *	6	17	35	−20	16.7	300–580	249/28N	22	48	−33	14.5
SBT15R *	3	21	204	−9	27.4	300–580	249/28N	247	204	+11	7.9
SBT15 *	3	67	6	−7	15.2	150–550	249/28N	63	11	−32	15.6
SBT16 *	3	302	25	−23	7.1	0–500	249/28N	356	39	−40	6.7
SBT17R *	6	24	180	−15	14.1	400–580	249/28N	24	180	+11	13.9
SBT17	6	28	120	−34	12.8	0–400	249/28N	42	127	−11	10.4
SBT18	6	14	141	−38	18.4	0–580	249/28N	14	145	−11	18.3
Rev	5	30	190	−5	14.1			39	30	−35	19.9
Normal	4	36	22	−16	15.6			17	193	+20	18.9
Mean	9	19	6	+3	12.3			16	20	−27	13.5
Pole Position: 69°S 181°E.											

* denotes sites used for pole position calculation. N = number of separately oriented samples. See Table 1 for other notation.

thermal demagnetisation up to 580°C (Fig. 7). Above 580°C less than 10% of magnetisation remained with the exception of sites SBT6 and 8 which did not demagnetise until 680°C. However, directions of components from SBT6 and 8 over 580–680°C are indistinguishable from those isolated below 580°C. Site mean component directions from the ten sites show a wide variation (Fig. 8; Table 2). Sites SBT10 and 17 contain two components of magnetisation (Table 2), a low-temperature (0–300°C) upward component with an easterly declination but markedly differing inclinations (intermediate component) and a higher temperature reversed-polarity component. Site SBT15 contains two approximately antiparallel components of magnetisation with a normal-polarity component in 3 samples and reversed-polarity in the remaining 3.

The magnetic mineralogy of the Paciencia Group has previously been described by Flint (1985b) who noted the occurrence of well rounded detrital grains of magnetite and rare titanomagnetite. Haematite is also present as an early authigenic cement and as an oxidation product of detrital magnetite, volcanic detritus and detrital micas. IRM acquisition curves (Fig. 7) and the absence of magnetisation above 580°C in 11 sites suggest that detrital magnetite is the primary remanence carrier in the Paciencia Group and that the contribution of haematite to the NRM is minimal.

The observation that detrital magnetite is the principal carrier of NRM in the Paciencia Group suggests that magnetisation was acquired during or shortly after deposition as either depositional remnant magnetisation (DRM) or post-depositional remnant magnetisation (PDRM). This interpretation is supported the polarity stratigraphy of the stable NRM components for the Artolla Member (Fig. 9). This shows that the site-mean directions define a reasonably clear polarity zonation. Site SBT15 which contains two approximately antiparallel components of NRM occurs on a polarity transition suggesting that sampling

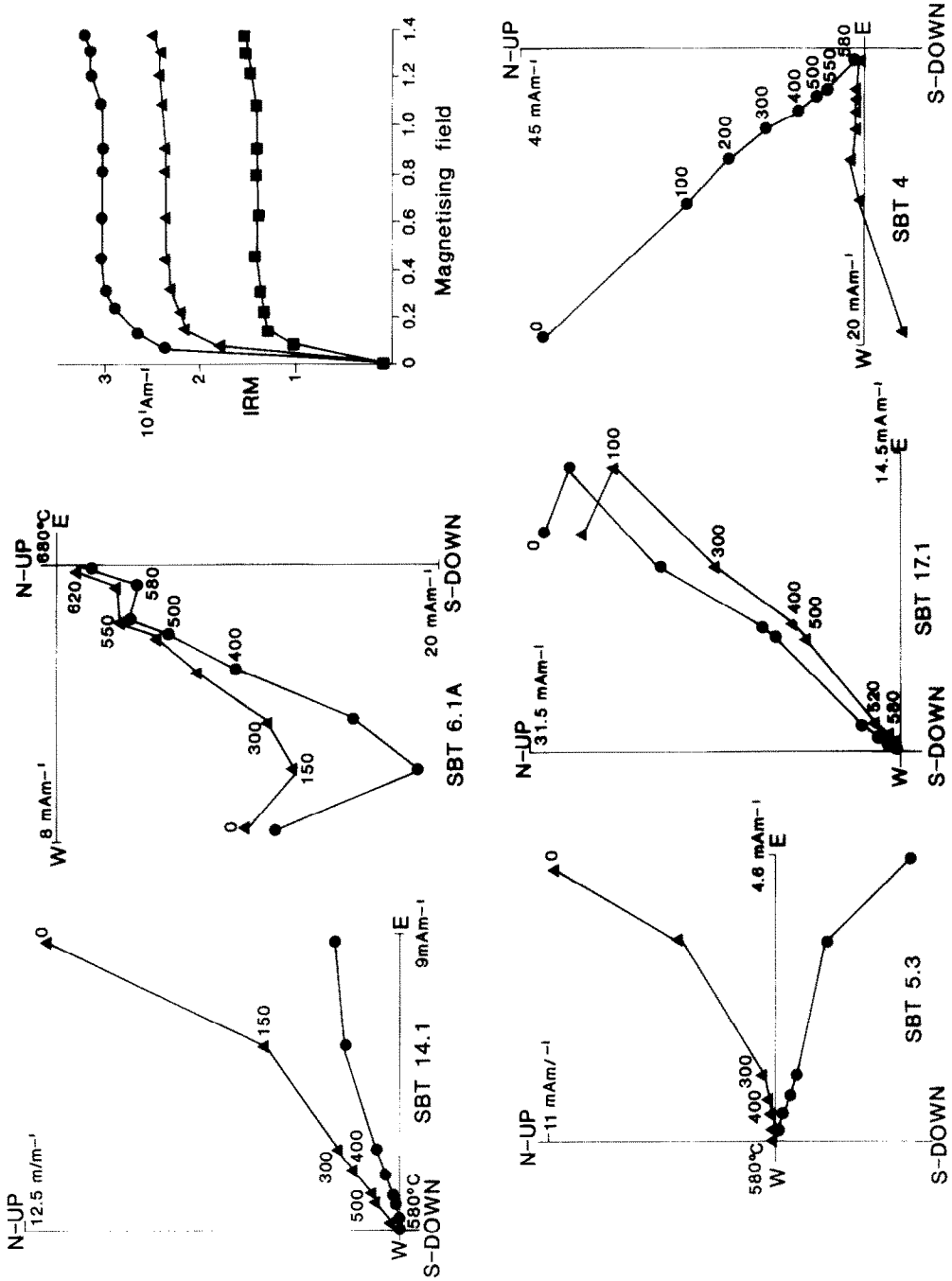


Fig. 7. Representative Zijderveld diagrams (circles = projection into horizontal plane, triangles = projection into vertical plane) and IRM acquisition curves for the Paciencia Group. Numbers adjacent to data points are thermal demagnetisation temperatures (°C).

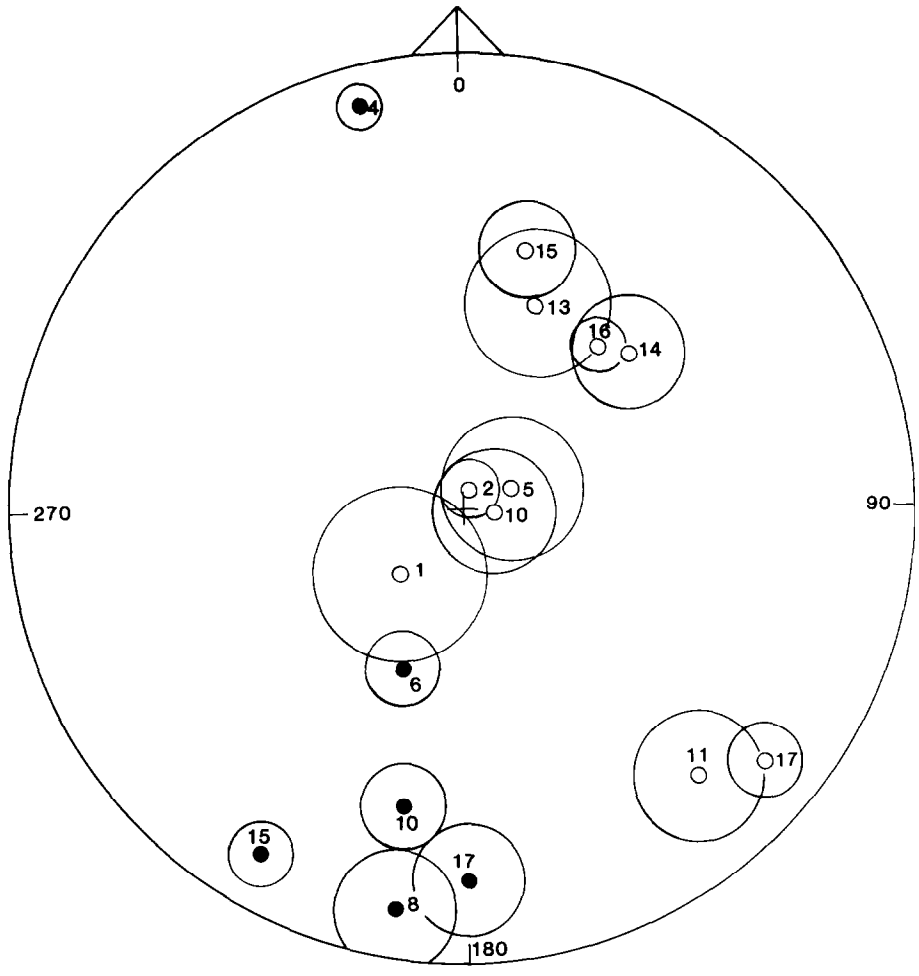


Fig. 8. Equal-area stereographic projection, site-mean NRM directions and α_{95} confidence limits for the Paciencia Group, keyed to Table 2. Open data points are in the upper hemisphere projection; filled data points are in the lower hemisphere.

straddled a transition. The above data indicate that the NRM of the lower part of the Paciencia Group is Oligocene in age.

Discussion

Butler et al. (1991) have recently determined a Late Cretaceous reference pole for South America, which is removed from the present-day rotation axis by 11° of arc. They concluded that significant apparent polar wander (APW) of the South American Plate must have occurred between 70 Ma and 10 Ma after which the palaeomagnetic pole is indistinguishable from the present rotation axis. The work of Butler et al. (1991) provides a suitable refence pole for the Purilactis Formation. However, Cenozoic reference poles have a marked scatter (Fig. 10) making reference pole selection for the Paciencia Group difficult (see Beck, 1988, and Butler et al., 1991, for discussion). The magnetisation of the Paciencia Group is probably Late Oligocene in age, and, whilst it is likely that an Oligocene reference pole for South America could be distinguished from the present rotation axis (see Butler et al., 1991), no such pole is available. Consequently, the present rotation axis is used as a reference pole (and given an arbitrary confidence limit of 5°), as, the time gap between the late Oligocene and the time at which the palaeomagnetic pole for South America becomes indistinguishable from the present rotation axis (i.e. between 20 and 10 Ma) is

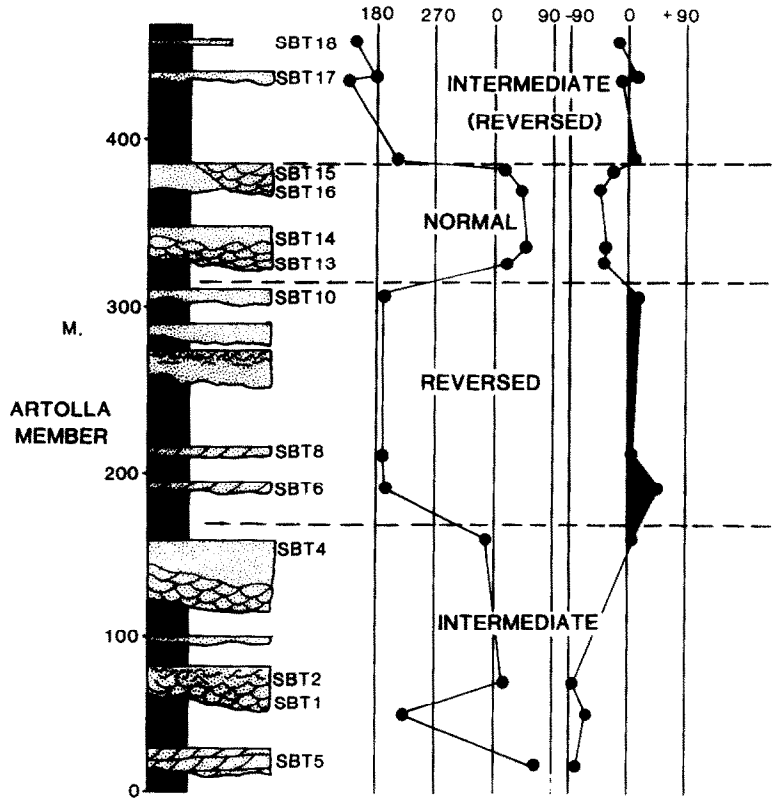


Fig. 9. Magnetic polarity zonation from the Artolla Member of the Paciencia Group.

less than the time gap between the Late Oligocene and the Late Cretaceous pole of Butler et al. (1991) (65 and 20 Ma).

Comparison of the Purilactis Formation and Paciencia Group poles with the late Cretaceous pole for cratonic South America and the present geographic pole respectively (Fig. 10; Table 3), indicates that substantial but significantly different amounts of clockwise rotation have affected the two units. Examination of structural, stratigraphic and palaeomagnetic data for the Precordillera suggests that the following series of events occurred after deposition of the Purilactis Formation:

(1) Magnetisation, uplift, folding and 12° of clockwise rotation of the Purilactis Formation prior to deposition of the Paciencia Group. This tectonic episode has been attributed to movement on the FDT and the WBT (Jolley et al., 1990; Fig. 4). The age of deformation is difficult to constrain owing to the continental nature of the two units. However, it is probably post-Palaeocene and pre-Oligocene and may correspond to the Lower Eocene Incaic Orogeny (Coira et al., 1982) which has a thin-skinned style of deformation in southern Peru (Mégard, 1984) similar to that of the Precordillera (Jolley et al., 1990).

(2) Development of an onlap unconformity between the Paciencia Group and the Purilactis Formation and deposition and magnetisation of the Paciencia Group in Oligocene–Lower Miocene times during a period of relative tectonic and magmatic quiescence.

(3) Paciencia Group sedimentation was terminated by further compressional tectonism which resulted in folding into a broad syncline, tightening of the Purilactis syncline, tilting of the Purilactis/Paciencia unconformity 12° to the east, local tilting of Paciencia Group sediments to vertical in the hangingwall of the FDT and 17° clockwise rotation of both the Purilactis Formation and the Paciencia Group. Again this tectonic episode can be attributed to further movement

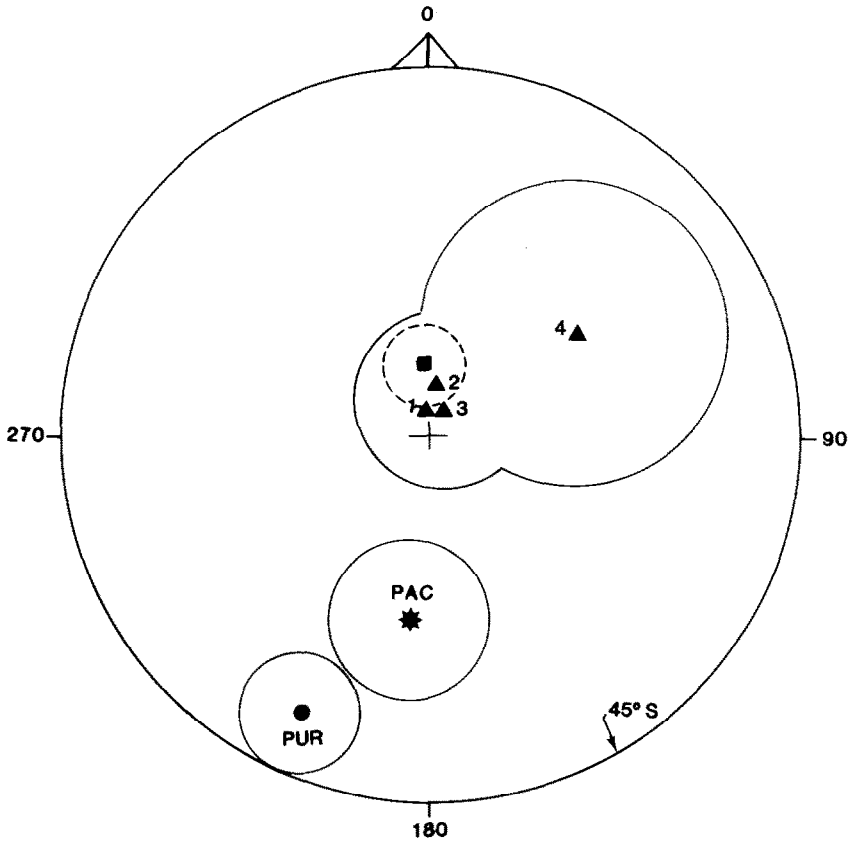


Fig. 10. Southern hemisphere plot showing the distribution of Cenozoic palaeomagnetic reference poles from South America (triangles) [(1) Irving and Irving (1982), (2) Valencio and Creer (1968), (3) Valencio et al. (1970) and (4) Schultz et al. (1986)], and the position of the Upper Cretaceous reference pole (square) determined by Butler et al. (1991), compared to the Paciencia Group (star) and the Purilactis Formation (circle) poles. Solid and dashed outlines denote 95% confidence limits.

on the FDT and WBT (Jolley et al., 1990). The age of deformation is thought to be Late Oligocene/Early Miocene and may correspond to the latest Oligocene/lower Miocene Peheunche tectonic phase recognised in southern Bolivia (Kussmaul et al., 1975) and northwest Ar-

TABLE 3

Rotation statistics for the Purilactis Formation (relative to the Upper Cretaceous pole of Butler et al., 1991) and the Paciencia Group (relative to the present-day rotation axis) using direction-space analysis

	Site location		Ex. Dir.		Observed		$R \pm \Delta R$	$F \pm \Delta F$
	Lat.	Long.	Dec	Inc	Dec	Inc		
Paciencia Group	22.6°S	68.3°W	3	−40	20	−27	17 ± 13	−13 ± 13
Purilactis Fmn.	22.8°S	68.4°W	12	−44	41	−36	29 ± 10	−8 ± 21

Statistics after Beck (1980) and Demarest (1983). Lat. = latitude of site sampled; Long. = longitude of site sampled; Ex. Dir. = expected direction of declination (Dec) and inclination (Inc) for the sampled area assuming a dipole magnetic field with pole at latitude 78.7°S and longitude 358.4°E (Butler et al., 1991) for the Purilactis Formation and the present-day pole position for the Paciencia Group; $R \pm \Delta R$ = Rotation, the difference between observed and expected declination with 95% confidence limits; $F \pm \Delta F$ = Flattening, the difference between expected and observed inclination with 95% confidence limits.

gentina (Salfity et al., 1984). This deformation may also correlate with the start of the main phase of uplift of the western flank of the Andean Altiplano (Mortimer and Saric, 1975).

Subsequent tectonic activity is associated with the eastward propagation of the thrust front into the Salar de Atacama to form the Cordillera de la Sal (Fig. 4) and took place in Pliocene and post-Quaternary times (Jolley et al., 1990). The WBT and FDT are thought to have been largely uninvolved in these tectonic events although minor post-Miocene rotation of the Purilactis Formation and Paciencia Group cannot be ruled out.

Implications for the Evolution of the Andean forearc

The above data indicate that rotation within the Andean forearc 250 km from the Peru-Chile trench is driven by a relatively thin-skinned compressional tectonic regime that was periodically active throughout the Cenozoic and probably exploited pre-existing faults. Palaeomagnetic data from the Cordillera de la Costa just south of Antofagasta (Fig. 1) suggest that fault-block rotation in post-Early Cretaceous times was the most important form of rotation, whether related to extension (Hartley et al., 1988a), right-lateral strike-slip (Beck, 1988) or left-lateral strike-slip (Armijo and Thiele, 1990). In contrast, 100 km south of the study area in the Precordillera, Jesinkey et al. (1987) showed that Upper Palaeozoic rocks had not undergone any rotation. Consequently within a relatively small area of the north Chilean forearc (approximately 1° of latitude) areas of non-rotation co-exist with areas of rotation. Areas which have been rotated display different structural styles (thin-skinned compressional or block fault rotation) suggesting that the type and timing(?) of rotation in the forearc is distinctly variable.

The palaeomagnetic data from the Chilean Precordillera also help to constrain the evolution of the Bolivian Orocline. Isacks (1988) recently showed that the development of the Bolivian Orocline took place in the lower/mid-Miocene and was characterised by thin-skinned and 'basin and range' style compressional deformation. Isacks also used palaeomagnetic data from the forearc to support this model. However, within the Precordillera it can be clearly demonstrated that rotation of the Purilactis Formation took place prior to the start of shortening related to Orocline development. Also, due to uncertainties in the timing and style of rotation of Mesozoic rocks from the forearc (many Mesozoic rocks have undergone at least two orogen-wide deformation events prior to the Early Miocene—the Eocene Incaic and mid-Late Cretaceous Peruvian Orogenies), it seemes unlikely that all rotation can be attributed to the Neogene development of the Bolivian Orocline. It is probable therefore that any rotation related to orocline development may well supplement pre-Neogene rotation, particularly as some areas of the forearc have undergone substantially more rotation (30–70°, Palmer et al., 1980; Heki et al., 1984; Forsythe et al., 1987; Roperch and Carlier, 1990) than the 10–20° rotation of each limb required to form the orocline.

In conclusion, any model for the evolution of the Andean forearc must take into consideration the different amounts, mechanisms and periods of rotation. First-order models previously presented to account for palaeomagnetic data from the Andean Orogen should be more rigourously constrained by structural data.

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